



Breathing Guide:

-Breathing Overview and Importance-

- Breathing is foundational for optimal movement, especially in the perinatal time frame
- Our core canister is made up of our diaphragm at the top, our pelvic floor muscles at the bottom, our abdominals in front and on the sides, and deep back muscles in back
 - We must be able to coordinate between all these muscles properly to effectively manage pressure, and it all starts with the diaphragm and breathing
- Diaphragmatic Breathing: Engages the diaphragm, supports core function, and reduces stress
 - The Diaphragm is a dome shaped muscle that sits at the base of the rib cage and is the primary muscle of breathing
 - Note: deep, diaphragmatic breathing is *not* the same as belly breathing. Our goal is 360 degree breathing—breathing and expanding into our back, sides, and belly
 - Decreased diaphragm expansion can result from:

Flared rib cage	Poor posture
Tense/overactive pelvic floor	Habitual shallow breathing pattern

-Breathing During the Perinatal Time Frame-

- During pregnancy:
 - The diaphragm gets pushed up and the ribs get pushed out, making it harder to take a deep breath in or a full exhale
 - Low back muscles and hip flexors can begin to compensate
- Postpartum:
 - Focusing on getting back to more normal breathing pattern after the above described changes
 - Working on pressure management once baby is no longer in the belly but everything is still lengthened and less stable
- Breathing impacts the entire body:
 - Core stability and pelvic floor: the diaphragm and pelvic floor muscles coordinate together with each breath which influences abdominal pressure, core stability, and pelvic floor tension
 - Pain and tension: chronic tension and pain in muscles like the neck, shoulders, low back, and pelvic region can be reduced with better breathing mechanics/less use of secondary muscles of breathing
 - Stress Response and Recovery: Breathing patterns directly affect the sympathetic nervous system (“fight or flight”) and parasympathetic nervous system (“rest and digest”). With a good diaphragmatic breath, the diaphragm descends and activates the vagus nerve, which helps turn on more of the parasympathetic system



-Breathing During Exercise-

- The main key is just to make sure you ARE breathing rather than breath holding
 - Breath holding can give us an illusion of stability, but holding our breath is not functional! We need to learn how to move, lift, stabilize all while breathing!
 - Sometimes breath holding can also cause undue pressure into the pelvic floor or front of the abdomen
 - If you're really having trouble breathing consistently and find yourself accidentally breath holding, one tip is to count your reps aloud, that way you physically cannot breath hold the whole time
- Generally, I recommend exhaling on the "harder" part of the movement or exertion and inhaling on the opposite portion of the movement
 - i.e. inhale as you lower down into a squat, exhale as you rise
 - i.e. exhale as you bend elbows for biceps curl, inhale as you control lowering the weights back down
- If possible, try to inhale through your nose ("sniff") and exhale through your mouth ("blow")
- I like doing a nice loud exhale when possible (sometimes feels awkward in a public gym but worth it in my opinion!)
 - Added benefit to breathing with exercise: a strong exhale can also help engage our core and allow us to generate more force through our muscles

-Breathing After Exercise-

- A lot of my workouts program in a "parasympathetic tap in" or focused breathing portion at the end
 - After we get our heart rates up and stress our system through a workout (good stress on purpose, but stress nonetheless), we want to try to get our heart rate down and calm down our bodies
 - Stress through exertion which brings on quicken breathing, elevated heart rate, etc activates more of our sympathetic nervous system (fight or flight) and diaphragmatic breathing with extended breath duration, especially focusing on a long exhale, can help us activate more of our parasympathetic nervous system (rest and digest)
 - Spending a few minutes in a restorative position doing focused breathing can help signal to our bodies that the workout/stress stimulus is over and it is time to tap into a more calm state

I hope this was a helpful overview. For more thorough guidance, structured exercise plans for wherever you are in your motherhood journey, and consistent education and empowerment from me, consider signing up for a subscription to The Murphy Method.

-Erin Murphy, PT, DPT